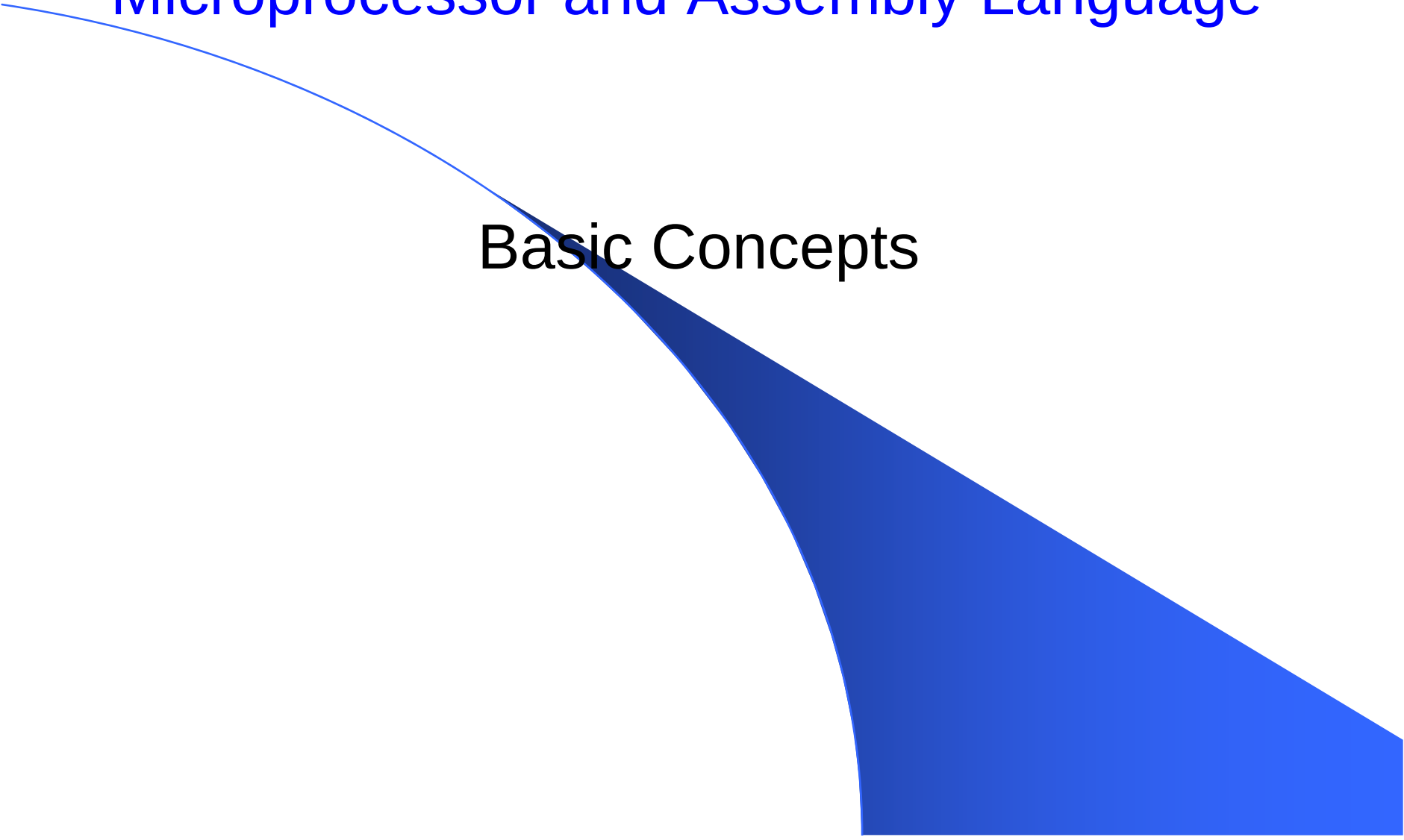


Microprocessor and Assembly Language

Basic Concepts

A decorative blue graphic is located in the lower right portion of the slide. It features a thin, light blue curved line that starts near the top left and sweeps downwards and to the right. Below this line, there is a solid blue shaded area that fills the space between the curve and the bottom right corner of the slide.

- Discipline (Listen Quietly, ask/speak one at a time)
- Mute MIC Raise hands
- Timing
- Ethics (Respect each others point of view)
- Try to contribute in order to reduce social problems/crimes
- Give way to others, don't block
- Course Material (WhatsApp Group by CR)
- Quizzes (4 or more)
- Assignments (Lab)
- Sessional/Final
- Course Outlines

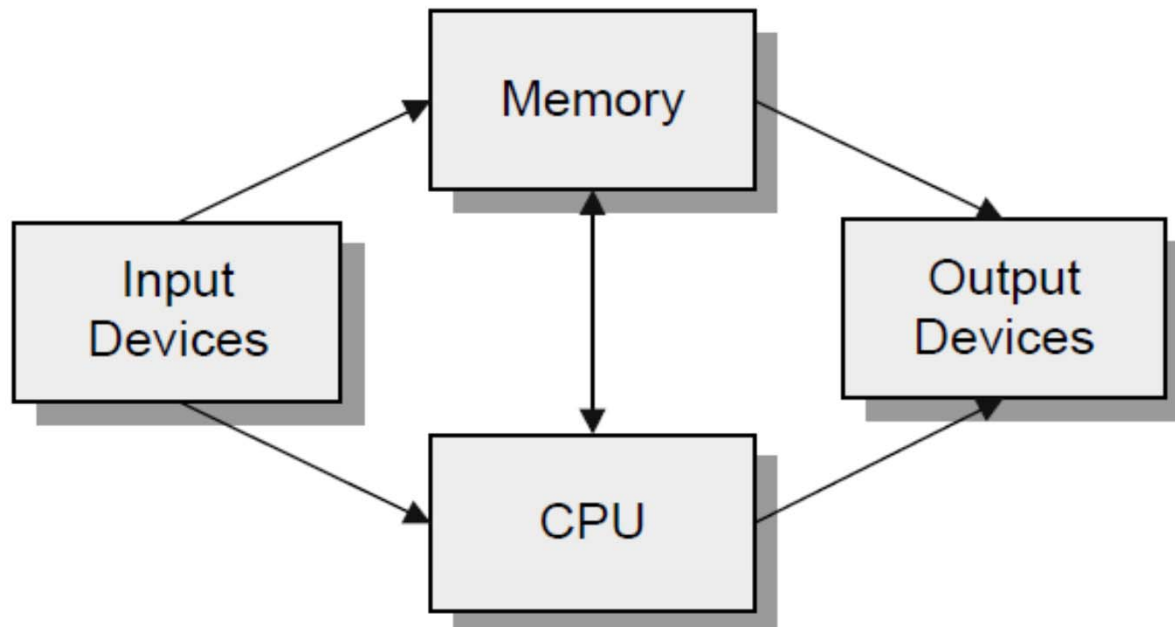
Recommended Books

- Assembly Language for Intel-Based Computers, 4th Edition by [Kip R. Irvine](#)
- Essentials of Computer Organization and Architecture by Linda Null and Julia Lobur ,4th Edition
- Computer Organization and Architecture, 6th Edition by William Stallings, Prentice Hall
- Computer Organization and Design, 4th Edition by David A. Patterson and John L. Hennessy

Computer Organization

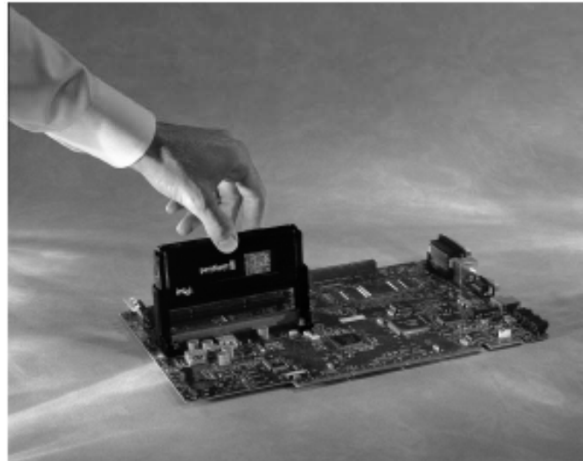
- ◆ CPU - central processing unit
 - Where decisions are made, computations are performed, and input/output requests are delegated
- ◆ Memory
 - Stores information being processed by the CPU
- ◆ Input devices
 - Allows people to supply information to computers
- ◆ Output devices
 - Allows people to receive information from computers

Computer Organization



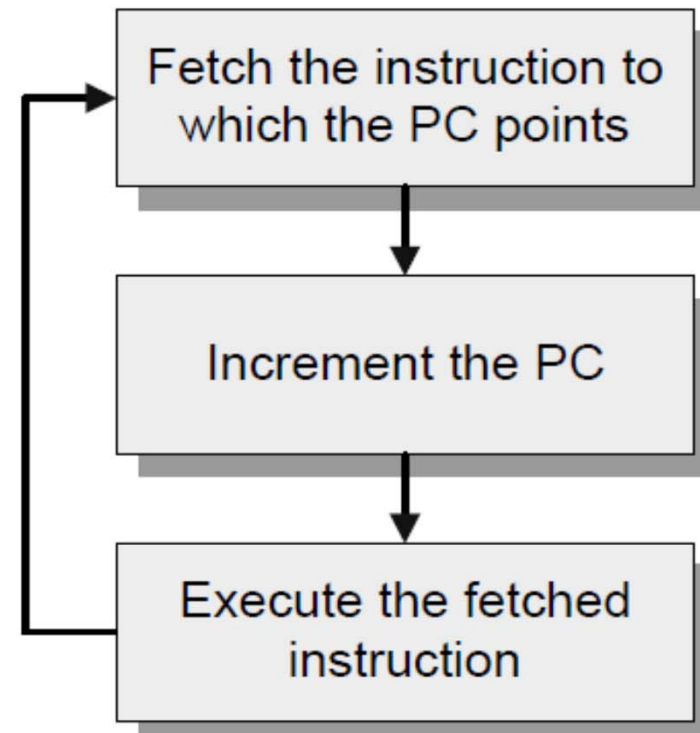
CPU

- ❖ *Brains* of the computer
 - Arithmetic calculations are performed using the Arithmetic/Logical Unit or ALU
 - Control unit decodes and executes instructions
- ❖ Arithmetic operations are performed using binary number system



Control Unit

- ◆ The fetch/execute cycle is the steps the CPU takes to execute an instruction
- ◆ Performing the action specified by an instruction is known as *executing the instruction*
- ◆ The program counter (PC) holds the memory address of the next instruction

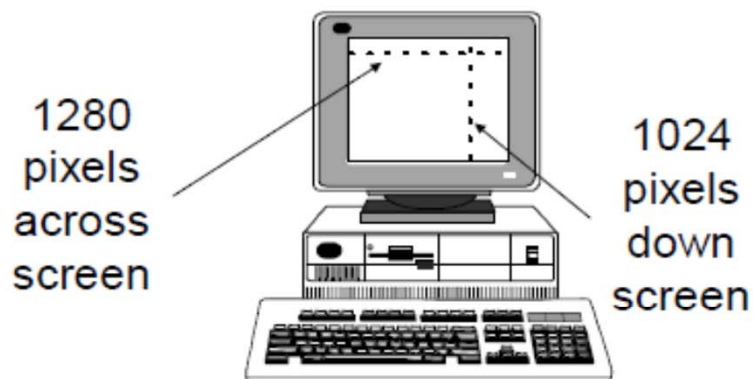


Input and Output Devices

- ◆ Accessories that allow computer to perform specific tasks
 - Receive information for processing
 - Return the results of processing
 - Store information
- ◆ Common input and output devices
 - Speakers Mouse Scanner
 - Printer Joystick CD-ROM
 - Keyboard Microphone DVD
- ◆ Some devices are capable of both input and output
 - Floppy drive Hard drive Magnetic tape units

Monitor

- ◆ Display device that operates like a television
 - Also known as CRT (cathode ray tube)
- ◆ Controlled by an output device called a *graphics card*
- ◆ Displayable area
 - Measured in dots per inch, dots are often referred to as pixels (short for picture element)
 - Standard resolution is 640 by 480
 - Many cards support resolution of 1280 by 1024 or better
 - Number of colors supported varies from 16 to billions



Software

- ◆ Application software
 - Programs designed to perform specific tasks that are transparent to the user

- ◆ System software
 - Programs that support the execution and development of other programs
 - Two major types
 - ◆ Operating systems
 - ◆ Translation systems

Application Software

- ◆ Application software is the software that has made using computers indispensable and popular
- ◆ Common application software
 - Word processors
 - Desktop publishing programs
 - Spreadsheets
 - Presentation managers
 - Drawing programs



Operating System

- ◆ Examples
 - Windows[®], UNIX[®], Mac OS X[®]
- ◆ Controls and manages the computing resources
- ◆ Important services that an operating system provides
 - File system
 - ◆ Directories, folders, files
 - Commands that allow for manipulation of the file system
 - ◆ Sort, delete, copy
 - Ability to perform input and output on a variety of devices
 - Management of the running systems